Proposed Research Direction

Evidence-aware uncertainty and selective responding in language-model systems

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STATUS

Early-stage research concept for discussion with potential supervisors. This is not a finalized proposal, validated novel method, completed experiment, or publication.

MOTIVATION

A language model may provide a fluent answer even when retrieved passages are incomplete, conflicting, outdated or only weakly relevant. In real decisions, the ability to identify unsupported claims and acknowledge uncertainty can matter as much as providing an answer. I am interested in how systems can make this decision more reliably.

RESEARCH QUESTION

How can an LLM-assisted system use evidence quality to decide when to answer, request further verification or abstain, and how can this decision be evaluated systematically?

INITIAL EXPERIMENTAL APPROACH

- Build test cases: Construct or adapt a small benchmark where questions are paired with sufficient, missing, ambiguous or contradictory evidence. Document ground-truth support and uncertainty conditions.
- Establish baselines: Compare an answer-always workflow, a prompt-only abstention approach and a basic retrieval/evidence-checking approach using the same questions and model settings.
- Evaluate decisions: Assess factual support, error types, appropriate abstention, calibration (if measurable), latency and computational overhead.
- Analyze limitations: Examine when evidence checks fail, whether results generalize across domains, and whether gains persist under harder cases.

EXPECTED RESEARCH LEARNING

The immediate goal is to learn rigorous evaluation and determine whether any improvement over strong baselines is reproducible. A potential thesis contribution would only be identified after literature review, supervisor feedback and preliminary experimental results.

BACKGROUND AND FIT

BS Information Technology, University of Agriculture Faisalabad, Sub-Campus Burewala (2025; CGPA 3.53/4.00). My final-year data-collection project and AquaSense AI learning notebook motivated my interest in data quality and reliability. I am currently strengthening Python, ML and experimental skills while working professionally in software development. I am seeking research-oriented Master's study in computer science or AI, with PhD research as a longer-term goal.

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